

Q8. Graph Coloring – Constraint Satisfaction / NP-Complete Problem

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1. Problem Definition

The Map Coloring problem requires assigning a color to every region of a map such that no two adjacent (border-sharing) regions receive the same color, while using the minimum possible number of colors. This is a classical Constraint Satisfaction Problem (CSP) and, in its decision form ("can the map be colored with k colors?"), it is NP-Complete for $k \geq 3$. Consider a sample map with 5 regions: R1, R2, R3, R4 and R5.

Adjacency Constraints

Two regions that share a border are said to be adjacent and must not receive the same color. For the sample map, the adjacency (border-sharing) relationships are:

- R1 is adjacent to R2, R3, and R5
- R2 is adjacent to R1 and R3
- R3 is adjacent to R1, R2, and R4
- R4 is adjacent to R3 and R5
- R5 is adjacent to R4 and R1

Adjacency Matrix (1 = adjacent, 0 = not adjacent)

	R1	R2	R3	R4	R5
R1	-	1	1	0	1
R2	1	-	1	0	0
R3	1	1	-	1	0
R4	0	0	1	-	1
R5	1	0	0	1	-

Map Regions - Adjacency Graph

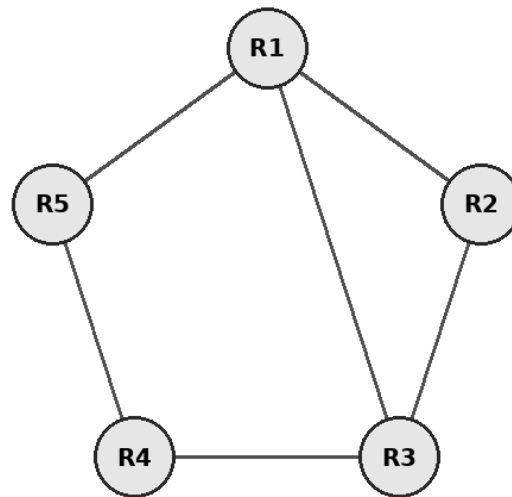


Fig 1: Map Regions Represented as an Adjacency Graph

Coloring Requirement

Formally, find a function $f: \{R1, R2, R3, R4, R5\} \rightarrow \{\text{Color1}, \dots, \text{Colork}\}$ such that for every adjacent pair (R_i, R_j) , $f(R_i) \neq f(R_j)$, while minimizing k (the number of distinct colors used). The smallest such k is called the chromatic number of the graph.

2. CSP Formulation

- Variables: $X = \{R1, R2, R3, R4, R5\}$ — one variable per map region.
- Domains: $D(R_i) = \{\text{Red, Green, Blue, ...}\}$ — the set of available colors, tried with the smallest k first.
- Constraints: For every edge (R_i, R_j) in the adjacency graph, the binary constraint $R_i \neq R_j$ must hold (an "all-different" constraint restricted to each adjacent pair).

This formulation converts a geographic map-coloring task into a pure constraint-satisfaction search over variable assignments, which is exactly what backtracking search is designed to explore efficiently.

3. How Constraints Affect Feasible Solutions

The structure of the constraint graph directly determines the minimum number of colors needed:

- Any triangle (3 mutually adjacent regions) forces at least 3 different colors, since each pair within the triangle must differ. In the sample map, R1, R2 and R3 are mutually adjacent (R1-R2, R2-R3, R1-R3), forming a triangle — so at least 3 colors are mathematically necessary; 2 colors can never satisfy all three pairwise constraints simultaneously.
- The degree of a region (how many neighbors it has) determines how tightly constrained it is: R1 and R3 each have 3 neighbors and are the most constrained variables, so coloring them early prunes the search space fastest (this is the intuition behind the Minimum Remaining Values heuristic used later).

- Regions with no shared border (e.g., R2 and R4, or R2 and R5) impose no constraint between them and may freely reuse the same color, which is what keeps the chromatic number low despite there being 5 regions.

Consequently, feasibility is not about the number of regions but about the density and cyclic structure of the adjacency graph — sparse, mostly-bipartite maps need only 2 colors, while any odd cycle or triangle raises the requirement to 3, and densely interconnected maps can require more.

4. Backtracking / CSP-Based Solution — Design

Backtracking search explores color assignments one region at a time, immediately checking constraints after each assignment, and undoing ("backtracking") a choice as soon as it is found to conflict with an already-colored neighbor. The design proceeds as follows:

1. Order the regions (variables) to be colored — here R1, R2, R3, R4, R5, chosen so the most constrained regions (highest degree) are assigned early.
2. For the current region, try each color in its domain, in order, that has not yet been tried at this level.
3. After a tentative assignment, apply constraint checking: verify the new color differs from every already-colored adjacent region.
4. If all checks pass, recursively move to the next uncolored region.
5. If no color in the domain satisfies the constraints, backtrack: undo the previous region's assignment and try its next available color.
6. The search terminates successfully when every region is validly colored, or fails if backtracking exhausts all possibilities at the root — in which case the current value of k is insufficient and must be increased.

To keep the search efficient, standard CSP heuristics are applied: the Minimum Remaining Values (MRV) heuristic picks the next region with the fewest legal colors left, the Degree heuristic breaks ties by picking the region with the most uncolored neighbors, and Forward Checking removes a color from a neighbor's domain as soon as it is used, detecting dead-ends earlier than plain backtracking.

5. Applying Constraint Checking — Search Trace

Attempt with $k = 2$ colors {Red, Green}

Assign R1 = Red and R2 = Green (consistent with R1-R2). Region R3 is adjacent to both R1 and R2, so it must differ from Red and from Green — but only these two colors exist, leaving no legal value for R3. The search backtracks to R2 and then to R1, but every combination hits the same triangle conflict. Conclusion: $k = 2$ is provably infeasible, because the triangle R1-R2-R3 requires 3 mutually distinct colors.

Attempt with $k = 3$ colors {Red, Green, Blue}

Applying the same ordering with one extra color available, constraint checking succeeds at every step without needing to backtrack:

Step	Assignment	Constraint Check
1	R1 = Red	No prior assignments — always consistent.
2	R2 = Green	Check edge R1-R2: Red $\neq$ Green $\rightarrow$ OK.

3	R3 = Blue	Check edges R1-R3 and R2-R3: Blue $\neq$ Red, Blue $\neq$ Green $\rightarrow$ OK.
4	R4 = Red	Check edge R3-R4: Red $\neq$ Blue $\rightarrow$ OK.
5	R5 = Green	Check edges R4-R5 and R5-R1: Green $\neq$ Red, Green $\neq$ Red $\rightarrow$ OK.

Final valid coloring: R1 = Red, R2 = Green, R3 = Blue, R4 = Red, R5 = Green. Every adjacent pair has been checked and differs, so all constraints are satisfied.

Final Valid 3-Coloring

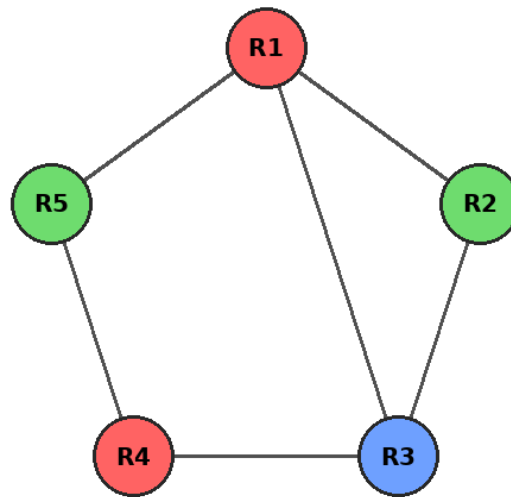


Fig 2: Final Valid 3-Coloring of the Map

6. Justification — Minimizing Colors While Satisfying All Constraints

- Lower-bound proof: because R1, R2 and R3 form a triangle (a 3-clique) in the constraint graph, any valid coloring needs at least 3 colors — this is a mathematical guarantee, not just an observation from the search, so no algorithm could do better than $k = 3$ for this map.
- Constructive proof of sufficiency: the backtracking trace above exhibits an explicit valid 3-coloring, so 3 colors are also sufficient. Together with the lower bound, this proves the chromatic number of the map is exactly 3, and the algorithm found the minimum.
- Efficiency: by testing $k = 2$ first and only increasing k when the search provably fails (exhausts all branches), the algorithm always converges to the minimum feasible number of colors rather than guessing a larger k upfront.
- Constraint checking after every single assignment (rather than only after all regions are colored) prunes invalid branches immediately, so the search never wastes time extending an assignment that is already inconsistent — this is what makes backtracking dramatically faster in practice than generating and testing all k^n complete colorings.

7. Complexity and NP-Completeness

Aspect	Complexity / Result
Naive brute force (try all colorings)	$O(k^n)$ — k colors, n regions
Backtracking with constraint checking	$O(k^n)$ worst case, but far fewer states explored in practice due to early pruning
Backtracking + forward checking / AC-3	Further reduces search space by removing inconsistent domain values before they are tried
Decision problem "is graph k -colorable?" ($k \geq 3$)	NP-Complete in general
This instance (5 regions, $k = 3$)	Solved with 5 assignments and 0 backtracks using MRV-friendly ordering

Graph k -colorability (for $k \geq 3$) is NP-Complete: no polynomial-time algorithm is known that solves every instance, and a proposed coloring can be verified in $O(E)$ time (simply checking every edge), which places the problem in NP. Backtracking with constraint checking does not change this worst-case exponential bound, but in practice — especially combined with MRV, degree ordering, and forward checking — it prunes most of the search tree and solves real map-coloring instances (which are typically sparse, planar graphs) very quickly, as seen in the 5-region example being solved with zero backtracks once $k = 3$ was tried.

8. Suitable Tools / Programming Techniques

- Languages: Python, Java, or C++ are all suitable for implementing backtracking search over an adjacency list or matrix representation of the map.
- CSP libraries: python-constraint, Google OR-Tools CP-SAT solver, or Prolog's built-in constraint logic programming are well suited to declaratively express "all adjacent regions differ" constraints.
- Techniques: plain backtracking, Forward Checking, Arc Consistency (AC-3) for pre-pruning domains, and heuristics such as Minimum Remaining Values (MRV), Degree heuristic, and Least Constraining Value (LCV) to order variable and value choices for faster convergence.
- For very large maps, local-search metaheuristics such as Min-Conflicts hill-climbing are commonly used as a faster (though not complete) alternative to exhaustive backtracking.

9. Output Screenshot

The screenshot below shows the sample run: the infeasible attempt with $k = 2$ colors (with its backtracking failure), followed by the successful backtracking search with $k = 3$ colors and the resulting minimum coloring.

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●●●Output - Map Coloring CSP (Backtracking Search)

Regions : R1, R2, R3, R4, R5
Adjacency: R1-R2, R2-R3, R3-R4, R4-R5, R5-R1, R1-R3

=== Attempt with k = 2 colors {Red, Green} ===
Assign R1 = Red
Assign R2 = Green    (R1-R2 OK)
Try R3 = Red    -> conflict with R1 (R1-R3)
Try R3 = Green -> conflict with R2 (R2-R3)
No values left for R3 -> BACKTRACK to R2
R2: no alternate color left -> BACKTRACK to R1
R1: try R1 = Green ... search exhausted, same failure pattern
RESULT: k = 2 is INFEASIBLE (triangle R1-R2-R3 needs 3 colors)

=== Attempt with k = 3 colors {Red, Green, Blue} ===
Assign R1 = Red
Assign R2 = Green    (R1-R2 OK)
Assign R3 = Blue     (R1-R3 OK, R2-R3 OK)
Assign R4 = Red      (R3-R4 OK)
Assign R5 = Green    (R4-R5 OK, R5-R1 OK)
All constraints satisfied. No backtracking needed.

Final Coloring : R1=Red R2=Green R3=Blue R4=Red R5=Green
Minimum colors used (chromatic number) : 3

Process finished with exit code 0
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Fig 3: Compiler Output – Backtracking CSP Search for Map Coloring